MINPPI · MULTI-OBJECTIVE FEATURE SELECTION

When average feature importance hides context-dependent harm

When MinPPI returns −1, it records a concrete coalition where adding one feature makes an acceptable model fail the utility quota; a prespecified validation rule decides whether to use the proposed removal set.

Wenjie SHI

Saint Petersburg State University

Applied Mathematics and Control Processes (AMCP)

st147487@student.spbu.ru

APPLICATION FOR PARTICIPATION
IN THE ASCOMP GRADUATE SCHOOL

THE RESEARCH QUESTION

Can a feature be harmful only in context, and can it be removed without consulting the test set?

STUDY SCOPE

Six retained benchmarks
Five frozen seeds
One model family

01 Why averages can miss harm

A feature can look harmless on average yet make one otherwise acceptable coalition fail the utility quota. MinPPI searches for that concrete counterexample.

ONE FOUND COALITION CERTIFIES HARM

ACCEPTABLE COALITION → ADD THE FEATURE → BELOW THE UTILITY THRESHOLD

The observed threshold flip is the witness: it explains exactly why the feature entered the removal candidate set.

EXACT FORM · BANZHAF REPRESENTATION

$$v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha U_{\text{full}}],\ U_{\text{full}} = U(\mathcal{M}_N),\ \alpha \in (0, 1]$$

$$\text{MinPPI}_i \equiv \text{MinBz}_i = \min_{S \subseteq N \setminus \{i\}} [v_\alpha(S \cup \{i\}) - v_\alpha(S)]$$

S = feature coalition · U = validation utility · α = quota · $\mathbb{1}[\cdot]$ = 1 if quota met, 0 otherwise

A found −1 certifies a threshold flip; sampled 0 or +1 may still miss a witness. Deletion requires the prespecified validation gate.

WHY THE DECISION IS AUDITABLE

- The witness is a concrete evaluated coalition.
- Harm is relative to the fixed model family, split, utility, and quota.
- A removal candidate is proposed, not automatically deleted.

02 How the decision is made

MinPPI separates finding evidence from deciding whether to act on it.

1. **Freeze the decision rule**
   Fix the model, split, utility, quota, and validation tolerance before selection.
2. **Find a harm witness**
   Find a coalition that meets the quota before feature i is added and falls below it afterwards.
3. **Apply the validation rule**
   Accept or veto the entire proposed removal set under the prespecified validation-utility tolerance.
4. **Open the held-out test last**
   Measure the frozen decision only after selection is complete.

SEPARATION OF ROLES

Witness search asks what can be harmful.
Validation asks whether removal is acceptable.
Held-out data asks what happened after the decision.

03 Results were selective and dataset-dependent

OVERALL PATTERN

Descriptively, average reductions were modest, while held-out utility stayed close to that of the full model. Effects nevertheless differed substantially by dataset.

FIVE-SEED CASE CONTRAST

	MEAN FEATURE-SET REDUCTION	HELD-OUT UTILITY
MADELON	44% fewer features	Held-out utility remained close to the full model.
ABALONE	28% fewer features	Held-out utility was lower than the full model.

Bars show five-seed mean reduction after the validation gate.

ELSEWHERE

Changes were small or mixed across the other four benchmarks.

SEARCH NOTE

Under the matched search budget, the clearest descriptive difference between prefiltered and full search appeared on Madelon. The pooled comparison was not statistically significant after correction, so recurrence elsewhere remains open.

THE PRACTICAL TAKEAWAY

An observed MinPPI value of −1 links a concrete threshold flip to an auditable removal candidate.

Feature removal is validation-gated and inspectable, not automatic.

WHERE THIS LEADS

- Adaptive witness search
- Broader replication across models and domains
- Stronger leakage screening

SCOPE & INTEGRITY Six retained benchmarks · five frozen seeds · one XGBoost model family.
Communities & Crime was excluded post hoc for outcome-component leakage; all aggregates were recomputed.